

Question 1 Random Forest: Mean Trait Model

This notebook evaluates the custom CSC 466 random forest using `data/processed/breed_traits_mean.csv`.

This version replaces the individual AKC trait ratings with grouped mean features: family life, physical, social, personality, and all-traits mean.

Model Interpretation

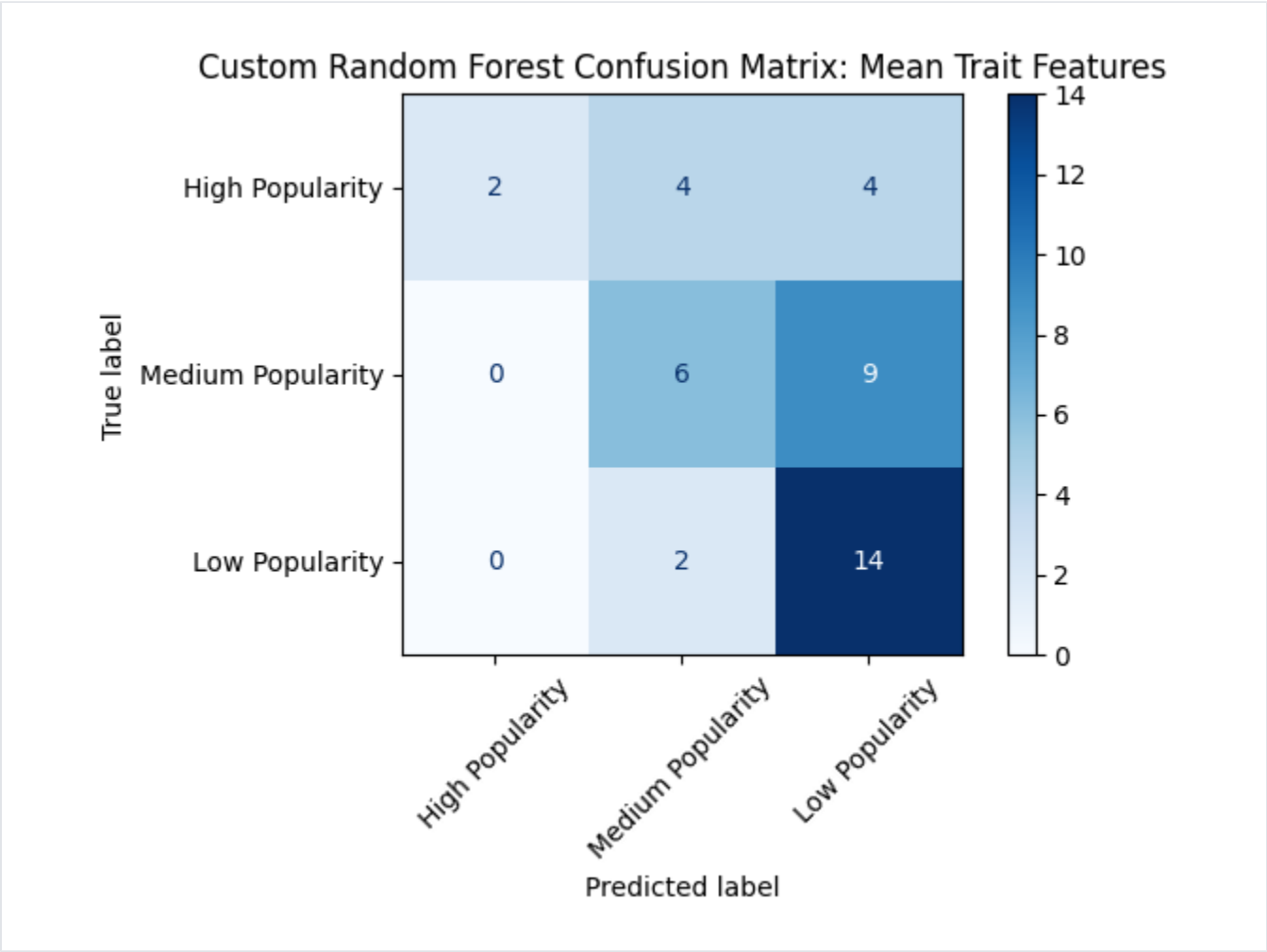
To interpret the model, these tables show which grouped features the forest split on most often and how average grouped trait values differ across the actual popularity tiers.

	Feature	Split Count
0	Personality	99
1	Social	91
2	Physical	78
3	Family Life	60

Popularity_Tier	High Popularity	Medium Popularity	Low Popularity
Family Life	4.278912	3.815789	3.939394
Physical	2.387755	2.267544	2.038961
Social	3.897959	3.707237	3.461039
Personality	3.750000	3.523026	3.597403

Popularity_Tier	Coat Type	Coat Length
High Popularity	Smooth	Short
Medium Popularity	Smooth	Medium
Low Popularity	Smooth	Short

	precision	recall	f1-score	support
High Popularity	1.00	0.20	0.33	10
Medium Popularity	0.50	0.40	0.44	15
Low Popularity	0.52	0.88	0.65	16
accuracy			0.54	41
macro avg	0.67	0.49	0.48	41
weighted avg	0.63	0.54	0.50	41



Prediction CSV

This output compares each test-set breed's actual popularity tier to the tier predicted by the mean-trait random forest model.

	Breed	Actual Popularity Tier	Predicted Popularity Tier	Correct
0	Doberman Pinscher	High Popularity	Medium Popularity	False
1	Dogue de Bordeaux	Medium Popularity	Medium Popularity	True

2	Sussex Spaniel	Low Popularity	Medium Popularity	False
3	Australian Cattle Dog	Medium Popularity	Low Popularity	False
4	Maltese	High Popularity	Low Popularity	False
5	Silky Terrier	Medium Popularity	Low Popularity	False
6	Yorkshire Terrier	High Popularity	Medium Popularity	False
7	Chinese Shar-Pei	Medium Popularity	Low Popularity	False
8	Lakeland Terrier	Low Popularity	Low Popularity	True
9	Akita	Medium Popularity	Medium Popularity	True
10	Irish Wolfhound	Medium Popularity	Low Popularity	False
11	Bloodhound	Medium Popularity	Low Popularity	False
12	English Springer Spaniel	High Popularity	Low Popularity	False
13	Greater Swiss Mountain Dog	Medium Popularity	Medium Popularity	True
14	Petit Basset Griffon Vendéen	Low Popularity	Medium Popularity	False
15	Greyhound	Low Popularity	Low Popularity	True
16	American Eskimo Dog	Medium Popularity	Low Popularity	False
17	Boston Terrier	High Popularity	Medium Popularity	False
18	Pug	High Popularity	High Popularity	True
19	Harrier	Low Popularity	Low Popularity	True
20	German Wirehaired Pointer	Medium Popularity	Low Popularity	False
21	Afghan Hound	Medium Popularity	Low Popularity	False
22	German Shepherd Dog	High Popularity	High Popularity	True
23	American Water Spaniel	Low Popularity	Low Popularity	True
24	Pumi	Low Popularity	Low Popularity	True
25	Manchester Terrier	Low Popularity	Low Popularity	True
26	Azawakh	Low Popularity	Low Popularity	True
27	Neapolitan Mastiff	Medium Popularity	Medium Popularity	True
28	Norfolk Terrier	Low Popularity	Low Popularity	True
29	Welsh Springer Spaniel	Low Popularity	Low Popularity	True
30	Chihuahua	High Popularity	Low Popularity	False
31	Bulldog	High Popularity	Medium Popularity	False
32	Bearded Collie	Low Popularity	Low Popularity	True
33	Shiba Inu	High Popularity	Low Popularity	False
34	Redbone Coonhound	Low Popularity	Low Popularity	True
35	American English Coonhound	Low Popularity	Low Popularity	True
36	Otterhound	Low Popularity	Low Popularity	True
37	Black Russian Terrier	Medium Popularity	Medium Popularity	True

38	Beauceron	Low Popularity	Low Popularity	True
39	Schipperke	Medium Popularity	Low Popularity	False
40	Leonberger	Medium Popularity	Medium Popularity	True